

Nesterov's Smooth Minimization of Non-Smooth Functions

Michał Szewczak Sebastian Trojan

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Based on Yurii Nesterov, *Smooth minimization of non-smooth functions*, 2005.

Motivation

Many convex optimization problems are non-smooth because their objectives contain maxima, norms, or absolute values.

Classical subgradient methods for non-smooth convex optimization have a worst-case rate of order

$$O(1/\varepsilon^2).$$

Nesterov's smoothing method uses additional max-structure of the objective. It replaces the non-smooth term by a smooth approximation and then applies an accelerated first-order method. The resulting theoretical complexity is improved to

$$O(1/\varepsilon).$$

The main trade-off is controlled by the smoothing parameter: smaller μ gives a more accurate approximation, but a less smooth problem.

Structured problem class

The original problem is

$$f^* = \min_{x \in Q_1} f(x),$$

where

$$f(x) = \hat{f}(x) + \max_{u \in Q_2} \{ \langle Ax, u \rangle - \hat{\phi}(u) \}.$$

- x is the primal variable.
- u is the auxiliary, or dual, variable.
- $\hat{f}(x)$ is the smooth part of the objective.
- $\hat{\phi}(u)$ is the part of the max-representation depending only on u .
- In the implemented examples, $\hat{f}(x) = 0$.

The non-smoothness comes from the maximization over u .

Smoothing the max-term

Choose a strongly convex prox-function d_2 on Q_2 , normalized so that

$$\min_{u \in Q_2} d_2(u) = 0, \quad D_2 = \max_{u \in Q_2} d_2(u).$$

Define the smoothed max-term

$$f_\mu(x) = \max_{u \in Q_2} \{ \langle Ax, u \rangle - \hat{\phi}(u) - \mu d_2(u) \}.$$

The full smoothed objective is

$$\bar{f}_\mu(x) = \hat{f}(x) + f_\mu(x).$$

The approximation error satisfies

$$\bar{f}_\mu(x) \leq f(x) \leq \bar{f}_\mu(x) + \mu D_2.$$

Therefore, choosing

$$\mu = \frac{\varepsilon}{2D_2}$$

makes the smoothing error at most $\varepsilon/2$.

Gradient and smoothness trade-off

The smoothed dual response is

$$u_\mu(x) = \arg \max_{u \in Q_2} \{ \langle Ax, u \rangle - \hat{\phi}(u) - \mu d_2(u) \}.$$

Then

$$\nabla f_\mu(x) = A^* u_\mu(x), \quad \nabla \bar{f}_\mu(x) = \nabla \hat{f}(x) + A^* u_\mu(x).$$

The gradient of $\bar{f}_\mu$ is Lipschitz continuous with constant

$$L_\mu = M + \frac{\|A\|_{1,2}^2}{\mu \sigma_2}.$$

Here M is the Lipschitz constant of $\nabla \hat{f}$, and $\|A\|_{1,2}$ is the operator norm associated with the chosen primal and dual geometries.

Decreasing μ improves approximation accuracy, but increases L_μ .

Accelerated minimization

For a fixed smoothing parameter μ , the method minimizes

$$\min_{x \in Q_1} \bar{f}_\mu(x).$$

One accelerated iteration

g_k	$= \nabla \bar{f}_\mu(x_k)$	gradient,
y_k	$= \arg \min_{y \in Q_1} \left\{ \langle g_k, y - x_k \rangle + \frac{L_\mu}{2} \ y - x_k\ ^2 \right\}$	local point,
G_k	$= \sum_{i=0}^k \alpha_i g_i, \quad \alpha_i = \frac{i+1}{2}$	accumulated gradient,
z_k	$= \arg \min_{x \in Q_1} \left\{ \frac{L_\mu}{\sigma_1} d_1(x) + \langle G_k, x \rangle \right\}$	aggregate point,
x_{k+1}	$= \frac{2}{k+3} z_k + \frac{k+1}{k+3} y_k$	next search point.

y_k uses the current gradient, while z_k uses all gradients accumulated in G_k .

Primal-dual certificate

The algorithm builds an averaged dual point

$$\hat{u} = \frac{\sum_{k=0}^N \alpha_k u_k}{\sum_{k=0}^N \alpha_k}, \quad u_k = u_\mu(x_k), \quad \hat{x} = x_N.$$

The dual lower-bound function is

$$\phi(u) = \min_{x \in Q_1} \{\hat{f}(x) + \langle Ax, u \rangle\} - \hat{\phi}(u).$$

By weak duality,

$$\phi(\hat{u}) \leq f^* \leq f(\hat{x}).$$

Therefore the primal-dual gap

$$f(\hat{x}) - \phi(\hat{u})$$

certifies the solution quality.

Theoretical iteration bound

Nesterov's estimate gives

$$f(\hat{x}) - \phi(\hat{u}) \leq \frac{4\|A\|_{1,2}}{N+1} \sqrt{\frac{D_1 D_2}{\sigma_1 \sigma_2}} + \frac{4MD_1}{\sigma_1(N+1)^2}.$$

In the implemented examples, $\hat{f} = 0$, so $M = 0$. Hence

$$f(\hat{x}) - \phi(\hat{u}) \leq \frac{4\|A\|_{1,2}}{N+1} \sqrt{\frac{D_1 D_2}{\sigma_1 \sigma_2}}.$$

To guarantee $f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon$, it is sufficient to take

$$N_{\text{pred}} + 1 = \left\lceil \frac{4\|A\|_{1,2}}{\varepsilon} \sqrt{\frac{D_1 D_2}{\sigma_1 \sigma_2}} \right\rceil.$$

Thus the predicted complexity is

$$N = O(1/\varepsilon).$$

Matrix game: $\min_{x \in \Delta_n} \max_{u \in \Delta_m} u^\top A x$

Continuous location: $\min_{\|x\|_2 \leq r} \sum_{j=1}^p w_j \|x - c_j\|_2$

Maximum of absolute values: $\min_{\|x\|_2 \leq r} \max_{1 \leq j \leq m} |a_j^\top x - b_j|$

Sum of absolute values: $\min_{\|x\|_2 \leq r} \sum_{j=1}^m |a_j^\top x - b_j|$

Experimental plan

- 1 Reproduce Nesterov's matrix game experiment.
- 2 Apply the same scheme to the other three examples.
- 3 Test the empirical dependence on ε .
- 4 Compare standard smoothing with a changing- μ strategy.

In every experiment, convergence is measured by the certified condition

$$f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon.$$

Experiment 1: matrix game reproduction

Reproduced problem:

$$\min_{x \in \Delta_n} \max_{u \in \Delta_m} u^\top A x, \quad A_{ij} \sim U[-1, 1].$$

Tested grid:

- $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\},$
- $m \in \{100, 300, 1000\},$
- $n \in \{100, 300, 1000, 3000, 10000\},$ with the largest n omitted for $\varepsilon = 10^{-4}.$

Each table cell reports iterations / percentage of the predicted iteration bound.

Experiment 1: matrix game results

$\varepsilon = 10^{-2}$					
$m \backslash n$	100	300	1 000	3 000	10 000
100	765/42%	934/46%	1 108/49%	1 187/49%	1 281/49%
300	690/34%	910/40%	1 126/45%	1 268/47%	1 475/51%
1 000	761/34%	873/35%	1 086/39%	1 284/43%	1 457/46%
$\varepsilon = 10^{-3}$					
$m \backslash n$	100	300	1 000	3 000	10 000
100	7 123/39%	8 426/41%	8 928/40%	10 191/42%	10 080/39%
300	7 337/36%	8 639/38%	10 154/40%	11 302/42%	12 022/41%
1 000	7 526/33%	8 828/35%	10 157/37%	11 595/39%	14 167/44%
$\varepsilon = 10^{-4}$					
$m \backslash n$	100	300	1 000	3 000	
100	68 505/37%	64 512/31%	77 136/34%	74 024/30%	
300	60 932/30%	78 680/34%	83 844/33%	92 862/34%	
1 000	70 778/31%	77 644/31%	93 346/34%	102 925/35%	

Experiment 2: other problem classes

The matrix game experiment was extended to:

- Continuous location:
 $\min_{\|x\|_2 \leq r} \sum_{j=1}^p w_j \|x - c_j\|_2, \quad c_j \sim \text{Unif}(B_2(0, 0.8r)), \quad w_j \sim U[0.5, 1.5], \quad r = 1.$
Grid: $p \in \{10, 50\}, d \in \{2, 5\}$ where $c_j, x \in \mathbb{R}^d$.
- Maximum of absolute values: $\min_{\|x\|_2 \leq r} \max_{1 \leq j \leq m} |a_j^\top x - b_j|, \quad A_{ji}, b_j \sim U[-1, 1], \quad r = 1.$
Grid: $m \in \{100, 300\}, n \in \{50, 200\}$ where $a_j, x \in \mathbb{R}^n$.
- Sum of absolute values: $\min_{\|x\|_2 \leq r} \sum_{j=1}^m |a_j^\top x - b_j|, \quad A_{ji}, b_j \sim U[-1, 1], \quad r = 1.$
Grid: $m \in \{100, 300\}, n \in \{50, 200\}$ where $a_j, x \in \mathbb{R}^n$.

Common accuracy levels:

$$\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}.$$

Experiment 2: summary tables

Continuous location				
ϵ	Avg. iter.	Max iter.	Avg. %	Time
10^{-4}	171 599	320 867	30.64%	31.8s
10^{-3}	20 525	31 653	43.74%	3.86s
10^{-2}	1 988	3 606	36.70%	0.35s
Maximum of absolute values				
ϵ	Avg. iter.	Max iter.	Avg. %	Time
10^{-4}	171 486	249 861	39.81%	10.6s
10^{-3}	20 032	33 671	44.76%	1.26s
10^{-2}	2 593	3 788	56.06%	0.16s
Sum of absolute values				
ϵ	Avg. iter.	Max iter.	Avg. %	Time
10^{-4}	1 198 404	2 711 368	29.45%	54.9s
10^{-3}	135 894	284 262	32.52%	6.43s
10^{-2}	12 088	27 750	30.43%	0.51s

Experiment 3: testing the $O(1/\varepsilon)$ behavior

To isolate the dependence on ε , five seeds were used; for each seed, one instance was generated and then kept fixed across all values of ε :

$$\varepsilon \in \{10^{-1}, 5 \cdot 10^{-2}, 10^{-2}, 5 \cdot 10^{-3}, 10^{-3}, 5 \cdot 10^{-4}, 10^{-4}\}.$$

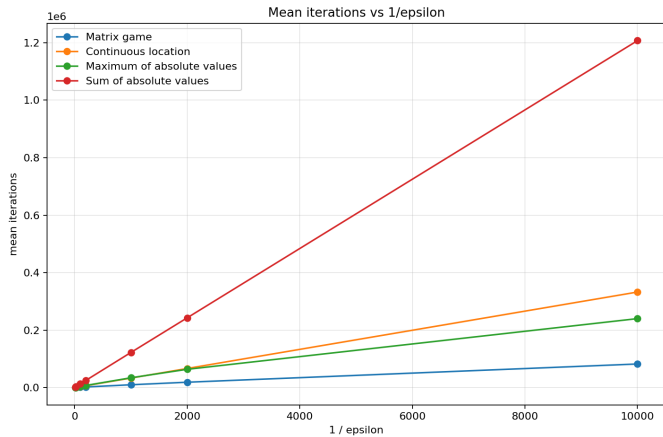
The theory predicts

$$N(\varepsilon) \approx C \frac{1}{\varepsilon}.$$

Therefore:

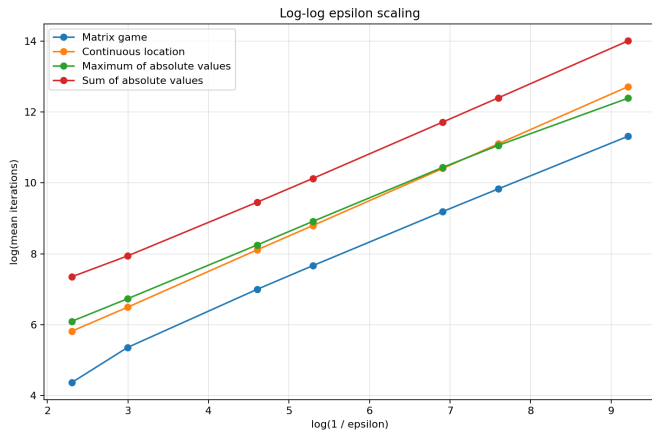
- iterations versus $1/\varepsilon$ should be approximately linear,
- the log-log fitted exponent $\hat{p}$ should be close to 1.

Experiment 3: iterations versus $1/\varepsilon$



The curves are close to linear; the difference between problems is mainly the constant factor.

Experiment 3: log-log scaling



Empirical power-law model:

$$N(\epsilon) \approx C\epsilon^{-p}.$$

Taking logarithms gives

$$\log N = \log C + p \log(1/\epsilon).$$

Therefore $\hat{p}$ is the slope of the fitted line on the log-log plot. The theory predicts

$$p = 1.$$

Problem	$\hat{p}$	SE
Matrix game	0.9897	0.0096
Continuous location	0.9986	0.0054
Max. abs. values	0.9208	0.0042
Sum abs. values	0.9649	0.0096

Experiment 4: changing the smoothing parameter

1. Motivation

- Standard smoothing uses $\mu_\varepsilon = \varepsilon/(2D_2)$ from the beginning.
- For small ε , this value is small, so the smoothed problem is sharper and harder.

2. Method

- Start with a larger smoothing parameter and decrease it geometrically:

$$\mu_0 = c\mu_\varepsilon, \quad \mu_{t+1} = \rho\mu_t.$$

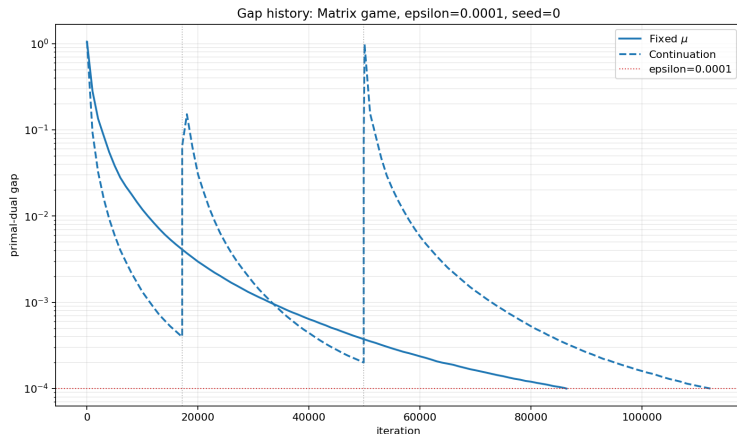
- Move to the next stage when

$$f(\hat{x}) - \phi(\hat{u}) \leq c_{\text{stage}}\mu_t D_2.$$

3. Our parameters

$$c = 8, \quad \rho = 0.5, \quad c_{\text{stage}} = 1, \quad 8\mu_\varepsilon \rightarrow 4\mu_\varepsilon \rightarrow 2\mu_\varepsilon \rightarrow \mu_\varepsilon.$$

Experiment 4: gap history



After reducing μ , the smoothed objective changes and the primal-dual gap can jump upward. In most tested cases, continuation required more iterations.

- The implementation reached the required primal-dual gap on all reported instances.
- Matrix game results reproduce the qualitative behavior from Nesterov's paper.
- The empirical scaling is consistent with the theoretical $O(1/\varepsilon)$ complexity.
- Worst-case iteration bounds were conservative on random instances.
- Practical difficulty depends strongly on the problem structure.
- The tested changing- μ strategy was usually less efficient than the standard smoothing method.

-  [Yurii Nesterov.](#)
Smooth minimization of non-smooth functions.
[Mathematical Programming](#), 103(1):127–152, 2005.